

TAPAS GUPTA

Principal Data Engineer / GenAI Architect / Cloud Data Platform

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PROFESSIONAL SUMMARY

Principal AI & Data Architect with 12+ years of experience designing enterprise-grade data platforms and productionizing Generative AI solutions. Expert in orchestrating RAG architectures, LLM workflows (LangChain, AWS Bedrock), and high-scale vector data processing. Proven track record of bridging the gap between complex data infrastructure and production-grade AI applications.

CORE SKILLS

- Cloud and Platforms: AWS, Databricks, S3, Lambda, API Gateway, Step Functions, EventBridge, DynamoDB, CloudFormation
- GenAI and AI Engineering: Bedrock, LangChain, RAG, prompt engineering, conversational AI, LLM workflows
- Data Engineering: Apache Airflow, Spark, PySpark, SQL, ETL / ELT, Delta Lake, data modeling, data quality
- Data Storage and Processing: S3, Postgres, DynamoDB, AWS OpenSearch Serverless, Pinecone
- DevOps and Automation: Docker, GitHub Actions, CloudFormation, Batch Job, Vault
- Messaging and Streaming: AWS Eventbridge, Kafka, RabbitMQ
- Governance and Cataloging: Atlan, Unity Catalog, metadata management, lineage, access controls
- Programming: Python, SQL, Flask

WORK EXPERIENCE

Globalization Partners / G-P - Principal Data Engineer/GenAI Architect

APR 2023 – DEC 2025

- Architected and deployed an enterprise-grade RAG (Retrieval-Augmented Generation) system via AWS Bedrock, leveraging specialized semantic chunking strategies and vector databases to enable contextual data retrieval from both S3 and Databricks tables.
- Designed and implemented automated LLM feedback loops, capturing implicit and explicit user signals (thumbs up/down, corrections) to continuously refine context chunking strategies and prompt alignment via reinforcement logging.
- Built an LLM-powered intelligent web-scraping solution using ChatGPT, Vertex AI, and prompt engineering to extract structured data from unstructured business sources.
- Enforced strict data governance, lineage, and row-level access controls (RBAC) across the AI retrieval layer by seamlessly integrating with the Unity Catalog and AWS ecosystem.
- Developed a web-scraping automation pipeline using Docker, AWS Batch, Step Functions, EventBridge, and Lambda for large-scale data collection.
- Designed and automated infrastructure and deployments using AWS CloudFormation, GitHub Actions, and monitoring tools, reducing release effort and improving operational reliability.
- Architected and delivered an enterprise data platform supporting multiple internal domains, enabling governed, scalable, and compliant data access.
- Developed internal serverless APIs using AWS Lambda, API Gateway, and DynamoDB to provide secure access for downstream teams.
- Integrated Atlan with the AWS ecosystem to improve metadata visibility, lineage tracking, and governance automation.
- Led technical talks, demos, and proof-of-concepts to accelerate GenAI adoption and platform maturity.
- Partnered with engineering, product, and governance stakeholders to onboard high-value datasets and define validation standards.

Zendesk – Senior Data Engineer

Jul 2021 – Mar 2023

- Executed CRM data migrations into Zendesk workspaces, ensuring seamless integration into the SaaS platform.
- Built data extraction and ingestion workflows from CSV, XML, web scraping, and third-party APIs including Salesforce, HappyFox, ConnectWise, Intercom, JIRA, Freshdesk, and ServiceNow.
- Developed and optimized an internal Python application using Flask, SQLAlchemy, Redis, Postgres, and Docker to improve data processing capabilities.
- Improved application performance by 30% by introducing multithreading, Celery, and Redis-based queue processing.
- Developed an AWS Lambda function to generate reports on unused EC2 resources across teams, significantly reducing annual AWS costs by automating bi-weekly reporting via AWS SES.

Tata Consultancy Services – Software Developer

Jan 2014–Jun 2021

- Designed and upgraded enterprise application integration systems using Python and Apache Airflow.
- Built ETL pipelines to ingest data from sources such as S3, MongoDB, and Snowflake.
- Writing REST web services for third-party tools for integrating into Airflow using Flask Framework.
- Containerized and deployed applications using Docker, Kubernetes, Helm, Vault, Sumo Logic, and Jenkins CI/CD.
- Scaled Airflow infrastructure using Celery executor, RabbitMQ, and Postgres.
- Wrote backend Python scripts to process large-scale data, perform data wrangling, and generate reports and charts.
- Automated EC2 instance configuration and launch processes using AWS CloudFormation, GitHub, and Docker, reducing manual effort by 20%.
- Mentored junior engineers and conducted peer reviews to improve delivery quality.

EDUCATION

NEW HORIZON COLLEGE OF ENGINEERING (VTU), BANGALORE, INDIA
Bachelor of Engineering in Computer Science

2009–13

CERTIFICATIONS AND AWARDS

- Google Vertex AI Agent Builder Training
- Innovation Pride Award for suggesting improvements in a client data platform
- Q1 Star Award for top performance